

Lockout/Tagout/Tryout (LoToTo) Procedure

ID#: **A-1107-MXZ01**
Created: 8/28/2024
Revised: 9/17/2024

Facility: **Trouw Nutrition Pvt Ltd** Location: **Production Tower 8th Floor**
India - Process Line
Description: **Bucket Elevator A-1107-MXZ01**

5

**Lockout
Points**

Note:

Turn off the Chain Conveyor and wait for the stopping sequence to be completed so that material in the Bucket Elevator gets discharged.

Electrical Main Power(EMP) should be Switch off When working on the Motor /Cable / Secondary isolation Point .

When in Doubt, Lock it Out !

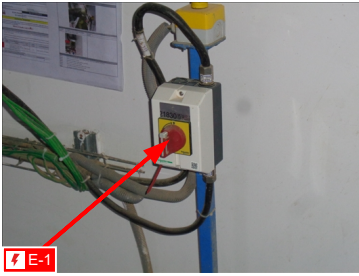
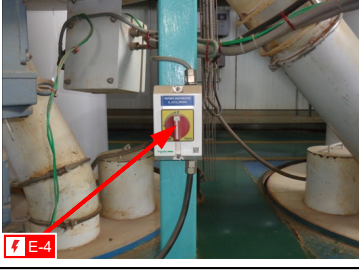
Purpose:	This procedure establishes the minimum requirements for the lockout of energy isolating devices whenever maintenance or servicing is done on machines or equipment.
Scope:	This procedure shall be used to ensure that the machine or equipment is stopped, isolated from all potentially hazardous energy sources and locked out before employees perform any servicing or maintenance where the unexpected energization or start-up of the machine or equipment or release of stored energy could cause injury.
Authorization:	This procedure shall only be used by employees that have been trained as "Authorized" employees. Employees with the following job roles are authorized to use this procedure.
Enforcement:	Failure to properly follow lockout-tagout procedure may result in corrective action.

Lockout Application Process

1. Stop all work equipment in the intervention area and notify all affected employees 2. Identify types and sources of hazardous energy. 3. Disconnect all hazardous energy sources. 4. Apply locks & tags. 5. Release and/or block any stored energy. 6. Verify zero energy.



Lockout Steps

Step #	Action	Info	Verification
1 Electrical  E-1 415 Volts Material Incomer Chain Conveyor Supply	The E-1 Isolator is located on the Bottom side of the machine at Under Ground Floor Marked has A-1106-MXZ01. Turn Isolator to the off position and lock out. Use a Lock device.		Verify zero voltage reading with meter. Note: Do try out to start the machine.
2 Electrical  E-2 415 Volts Bucket Elevator Star Supply	The E-2 Isolator is located on the Front side of the machine at 8th Floor Marked as A-1107-MXZ01. Turn Isolator to the off position and lock out. Use a Lock device.		Verify zero voltage reading with meter. Note: Do try out to start the machine.
3 Electrical  E-3 415 Volts Bucket Elevator Delta Supply	The E-3 Isolator is located on the Front side of the machine at 8th Floor Marked as A-1107-MXZ01. Turn Isolator to the off position and lock out.		Verify zero voltage reading with meter. Note: Do try out to start the machine.
4 Electrical  E-4 415 Volts Roatry Distributor Supply	The E-4 Isolator is located on the Bottom side of the machine at 8th Floor Marked as A-1111-MXZ01. Turn Isolator to the off position and lock out.		Verify zero voltage reading with meter. Note: Do try out to start the machine.
5 Electrical Main Power  EMP-1 415 Volts Main Supply	The EMP-1 Isolator is located on the Front side of the MCC Panel Marked 5F3 Star Delta Starter 1107-Bucket Elevator. Turn Isolator to the off position and lock out. Use a Lock device.		Verify there is no indication on panel. Note:- EMP should be off only During the Motor or Cable Change by Authorized person only.

Verification of Energy Isolation

Verify that all energy sources are isolated and at a Zero Energy State by attempting to start machine with normal operating controls.

Lockout Removal Process

1. Remove tools & loose items. 2. Ensure safeguards are back in place, secure, and the work equipment is OFF. 3. Confirm all personnel are clear. 4. Remove locks & tags. 5. Reconnect energy sources. 6. Turn ON and verify safe operation.

Lockout Tagout Procedure

Purpose:	To protect authorized employees against unexpected or unplanned activation of equipment or energy while servicing equipment.
Scope:	Utilize this procedure for all scheduled PM shutdowns, any maintenance task that requires you to place your body in harms way of the equipment, or if you have to leave the area while the equipment is in service.
Enforcement:	Failure to properly follow lockout-tagout procedure may result in corrective action.

SHUTDOWN, LOCK, TAG & TEST SEQUENCE

#	STEP	DESCRIPTION
1	<i>Notify Employees</i>	Notify all affected employees that servicing or maintenance is required on a machine or equipment, and that the machine or equipment must be shut down and locked out to perform the servicing or maintenance.
2	<i>Review Lockout Procedure</i>	The authorized employee shall refer to the company procedure to identify the type and magnitude of the energy that the machine or equipment utilizes, shall understand the hazards of the energy, and shall know the methods to control the energy.
3	<i>Perform Machine Stop</i>	If the machine or equipment is operating, shut it down by the normal stopping procedure (depress the stop button, open switch, close valve, etc.). Reference machine operating procedure for normal shutdown.
4	<i>Isolate Energy</i>	Follow graphical lockout-tagout procedure from top to bottom to de-activate the energy isolating device(s) so that the machine or equipment is isolated from the energy source(s). NOTE: It may be necessary to dissipate the non-lockable energy sources before isolating the lockable energy sources. (i.e. lower the machine to lowest position before locking out.)
5	<i>Lockout Energy</i>	Lock out and tag out the energy-isolating device(s) with assigned lock(s) and tag(s). If the lock(s) need to be transferred to another employee, follow the company procedure for authorized employee transfer.
6	<i>Dissipate Energy</i>	Stored or residual energy (such as that in capacitors, springs, elevated machine members, rotating flywheels, hydraulic systems, as well as air, gas, steam, or water pressure, etc.) must be dissipated or restrained by methods such as grounding, repositioning, blocking, bleeding down, etc.
7	<i>Attempt Restart</i>	Ensure that the equipment is disconnected from the energy sources by first checking that no personnel are exposed, then verify the isolation of the equipment by operating the push button or other normal operating controls or by testing to make certain the equipment will not operate. Caution: Return operating controls to neutral or "off" position after verifying the isolation of the equipment.

RESTORE TO SERVICE SEQUENCE

#	STEP	DESCRIPTION
1	<i>Check Machine</i>	Check the machine or equipment and the immediate area around the machine to ensure that nonessential items have been removed and that the machine or equipment components are operationally intact.
2	<i>Check Area</i>	Check the work area to ensure that all employees have been safely positioned or removed from the area.
3	<i>Verify Machine</i>	Verify that the controls are in neutral.
4	<i>Remove Lockout</i>	Remove the locks, tags and lockout devices and re-energize the machine or equipment. In reverse order, follow all of the steps from the visual lockout-tagout procedure found on the previous page. Note: The removal of some forms of blocking may require re-energization of the machine before safe removal.
5	<i>Notify Employees</i>	Notify affected employees that the servicing or maintenance is completed and the machine or equipment is ready for use.